



Society of Petroleum Engineers

SPE-226892-MS

Casing Reamer Shoe Powered by Positive Displacement Motor Enables Hard Reaming Casing String Through Extended Open-Hole Sections

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/226892-MS](https://doi.org/10.2118/226892-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

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Abstract

In extended open-hole sections, conventional casing deployment often encounters significant challenges, including tight spots, swelling formations, ledges, and accumulated cuttings. These conditions necessitate wiper trips or rotation of the casing string, increasing the risk of casing fatigue, high surface torque, and non-productive time (NPT). To address these issues, a motorized casing reamer shoe powered by a positive displacement motor (CRSPDPM) technology has been developed and utilized to hard ream the casing string. The CRSPDPM delivers localized high-torque, low-speed rotation at the casing shoe level to mechanically ream through obstructions without requiring full-string rotation. This innovative technology has been tested in both laboratory and field settings across challenging wells, demonstrating its ability to maintain flow paths even in the presence of lost circulation material (LCM), deliver torque outputs exceeding 5,200 ft-lbf, and reduce casing drag and deployment time significantly. This manuscript provides a comprehensive overview of the tool's design, operation principles, performance testing, and economic implications of its field deployment in deviated wells. The implementation of the CRSPDPM has improved operational efficiency, lowered casing failure risks, and ensured reliable bottomhole casing placement in extended lateral environments.

Introduction

Extended reach drilling (ERD) and horizontal development wells require reliable casing running technologies to ensure safe delivery of casing strings to total depth (TD). However, open-hole wellbores, particularly those through reactive shales, unconsolidated sandstones, and fractured intervals, are prone to collapse, tight hole conditions, and ledge formation. These conditions increase frictional drag and lead to excessive torque while running casing, making it challenging to achieve successful casing deployment. Historically, operators have addressed these issues with wiper trips, reaming-while-drilling (RWD), or rotating casing strings using top drives. While these approaches have been effective, they increase operational risk, add time, and accelerate fatigue on premium connections. Casing-while-drilling level-1 (CwD) technology has been utilized to hard ream casing strings across entire open-hole sections to total depth. Although it is an effective technology, it has its limitations, especially when there is a need to rotate

the entire casing string. The casing string is exposed to high torque, and it is exposed to fatigue risk if the wellbore has micro-doglegs. To mitigate these risks, a more efficient and reliable solution is required. The oil and gas industry has been seeking innovative solutions to address the challenges associated with casing deployment in extended open-hole sections. The development of new technologies, such as the CRSPDPM, has been driven by the need to improve operational efficiency, reduce costs, and minimize the risks associated with conventional casing deployment methods. The CRSPDPM technology has been designed to provide a reliable and efficient solution for casing deployment in extended open-hole sections, and its implementation has shown promising results in reducing the risks associated with conventional casing deployment methods.

Challenges with Conventional Casing Deployment

Conventional casing deployment methods often result in significant challenges, including high torque, casing fatigue, and non-productive time. The planning of 9-5/8-in casing strings, for example, requires careful consideration of the wellbore geometry, formation properties, and casing design. Engineering simulations are performed to confirm that applying casing drilling is technically feasible and that the torque and drag, stress and fatigue analysis are within the technical acceptable range. However, even with careful planning, the actual survey results may indicate an alarming torque range due to total losses and expected junk in the hole. Close monitoring of the torque during reaming is critical to ensure that no limits are crossed, which could lead to casing connection fatigue and twist-off. In some cases, the casing string may need to be upgraded to a more expensive connection type to withstand the expected service torque. These high-torque connection types are normally unusual requirements, which might increase the cost significantly. Furthermore, the use of conventional casing deployment methods can result in significant non-productive time, which can have a major impact on the overall cost of the well. The oil and gas industry has been seeking innovative solutions to address the challenges associated with casing deployment in extended open-hole sections. The development of new technologies, such as the CRSPDPM, has been driven by the need to improve operational efficiency, reduce costs, and minimize the risks associated with conventional casing deployment methods. The CRSPDPM technology has been designed to provide a reliable and efficient solution for casing deployment in extended open-hole sections, and its implementation has shown promising results in reducing the risks associated with conventional casing deployment methods.

Limitations of Existing Technologies

Existing technologies, such as casing-while-drilling level-1 (CwD), have limitations when it comes to rotating the entire casing string. The expected surface torque for the level-1 9-5/8-in CwD through an extended highly deviated wellbore (S-shape) can reach 40,000 ft-lbs, which is beyond the capability of most premium connections in the market. To overcome these challenges without compromising casing integrity or prolonging operations, a new technology is required. The CRMPPDPM technology has been developed as an alternative option, incorporating a positive displacement motor and reaming cutters. This technology reams with motor power any tight spots or collapsed walls encountered while casing running without the need to rotate the casing string. The CRMPPDPM technology has been designed to provide a reliable and efficient solution for casing deployment in extended open-hole sections, and its implementation has shown promising results in reducing the risks associated with conventional casing deployment methods. Furthermore, the technology has been engineered to be compatible with a wide range of wellbore conditions, including high-pressure and high-temperature environments. The CRMPPDPM technology has a much higher torque value due to the drilling mud motor working principle in comparison to existing turbine or casing reciprocating principles. The tool has a fixed output torque value that can be pre-agreed depending on the power section length. The CRMPPDPM also has a relatively high compatibility with loss circulation materials (LCM) that might be used during casing/liner running with losses scenarios. Additionally, the technology has been

designed to minimize the risk of turbine power section plugging or accidental sidetrack due to low-medium nose (bit) RPM.

CRMPPDM Technology

The CRMPPDM technology is a drillable tool that incorporates a positive displacement motor and reaming cutters. This technology has a much higher torque value due to the drilling mud motor working principle in comparison to existing turbine or casing reciprocating principles. The tool has a fixed output torque value that can be pre-agreed depending on the power section length. The CRMPPDM also has a relatively high compatibility with loss circulation materials (LCM) that might be used during casing/liner running with losses scenarios. Furthermore, in comparison to existing turbine rotatable reamers, there is no risk of turbine power section plugging or accidental sidetrack due to low-medium nose (bit) RPM. The CRMPPDM technology has been designed to provide a reliable and efficient solution for casing deployment in extended open-hole sections, and its implementation has shown promising results in reducing the risks associated with conventional casing deployment methods. The technology has been engineered to be compatible with a wide range of wellbore conditions, including high-pressure and high-temperature environments. The CRMPPDM technology has been tested in both laboratory and field settings, demonstrating its ability to maintain flow paths even in the presence of lost circulation material (LCM), deliver torque outputs exceeding 5,200 Ft-lbf, and reduce casing drag and deployment time significantly. The tool has been designed to be easy to operate and maintain, with a simple and intuitive interface. The CRMPPDM technology provides a reliable and efficient solution for casing deployment in challenging wellbores.



Figure 1—The 9-5/8-in CRMPPDM tool overview. The motor power section before installation is also shown

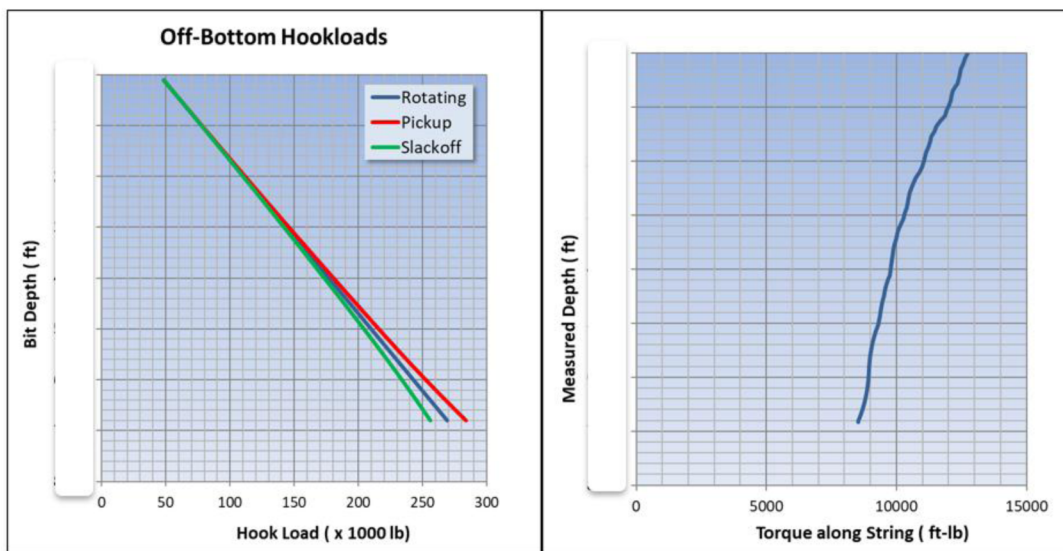


Figure 2—Planned torque & drag graphs simulation for a level-1 9-5/8-in CwD job through a short wellbore section. Through longer wellbores, the expected torque on the casing string is higher.

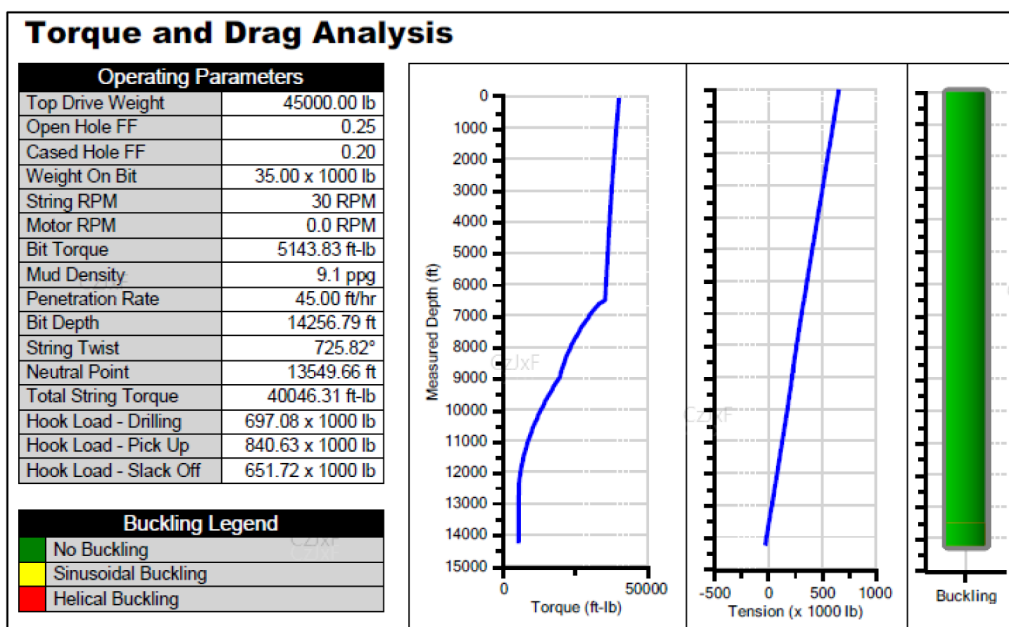


Figure 3—Planned torque & drag graphs simulation for a level-1 9-5/8-in CwD job through an extended highly deviated wellbore section. The expected surface torque while reaming the casing string reached 40 kft-lbs.

Technology Overview

Design Architecture

The CRSPDPM tool consists of a high-torque PDM integrated into a drillable shoe body with a reinforced reaming section. The reamer head features PDC and tungsten carbide cutters arranged for bidirectional reaming. Multiple nozzles—both sideward and downward—are engineered for hole cleaning and jetting. The tool functions at 95–792 GPM flow rates, producing rotational speeds between 50–150 downhole RPM rotary sppd, and torque outputs up to 5,200 ft-lbf.

The housing is designed for 9-5/8-in casing strings and is compatible with both OBM and WBM systems. The entire assembly, including the power section and nozzles, is constructed from corrosion-resistant duralumin and elastomeric rotor materials to withstand high-temperature conditions up to 150°C (302°F).

Cementing Safety Sub

The CRMPPDM has a cementing safety sub (figure 3a) designed to enable completion of the cementing operation through bypass ports if the reamer system got plugged while running the casing/liner. The optional cementing safety sub ensures cement placement through backup ports. Its sliding-sleeve mechanism is activated by pre-determined differential pressure (120–160 bar) and it is adjustable, controlled by the amount of shear pins installed. It also incorporates a rotor-locking mechanism activated by weight-on-bit (WOB) during drill-out to avoid tool spinning and ensure bit stability.

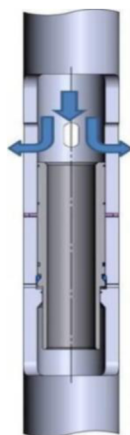


Figure 3a—The cementing safety sub and the flow path through the backup ports.

Laboratory Testing and Validation

Extensive laboratory tests were conducted to validate tool performance under simulated downhole conditions. A flow loop was constructed with a 9-5/8-in casing section and synthetic cuttings. Flow tests at various rates confirmed consistent RPM-torque profiles across pressure ranges as shown in figure 4.

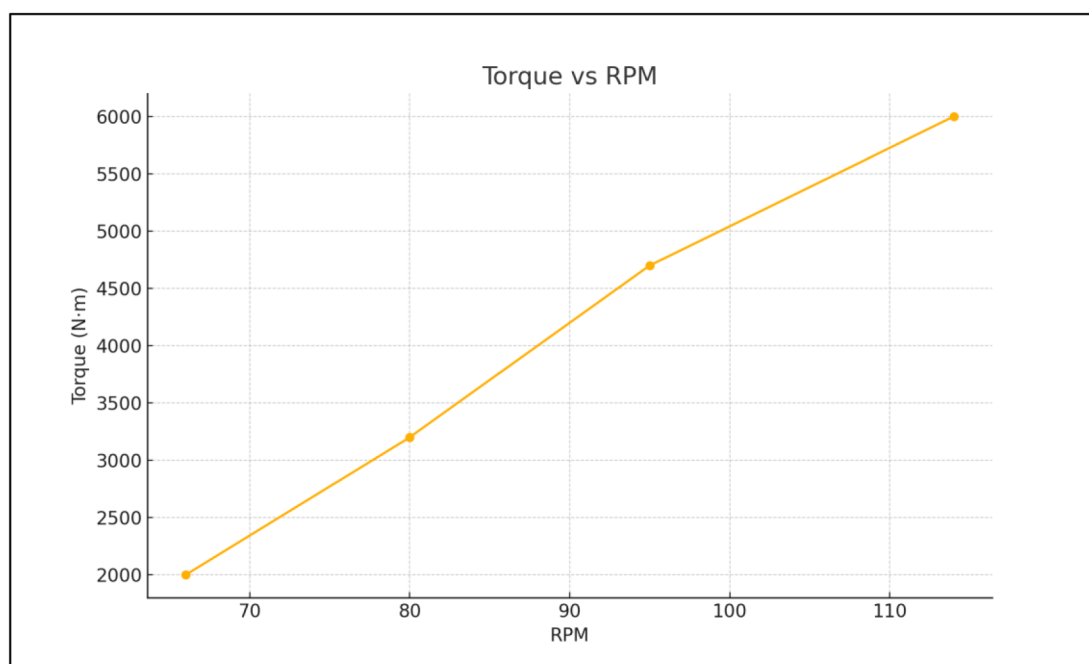


Figure 4—Torque vs rotary speed relationship between tool RPM and output torque under standard flow conditions.

Cuttings beds, simulated tight spots, and elastomeric materials were introduced to test reaming cutter durability. Torque readings showed a linear increase with flow-induced rotor speed. At peak flow (3,000 LPM), the tool maintained 6,000 N·m torque output with no loss in rotational stability. Tool flow paths were subjected to 45 lb/bbl LCM concentration exposure with zero plugging.

Durability testing included 150-hour continuous runtime with thermal cycling. Cutter integrity remained intact. No degradation in elastomeric stator was recorded.

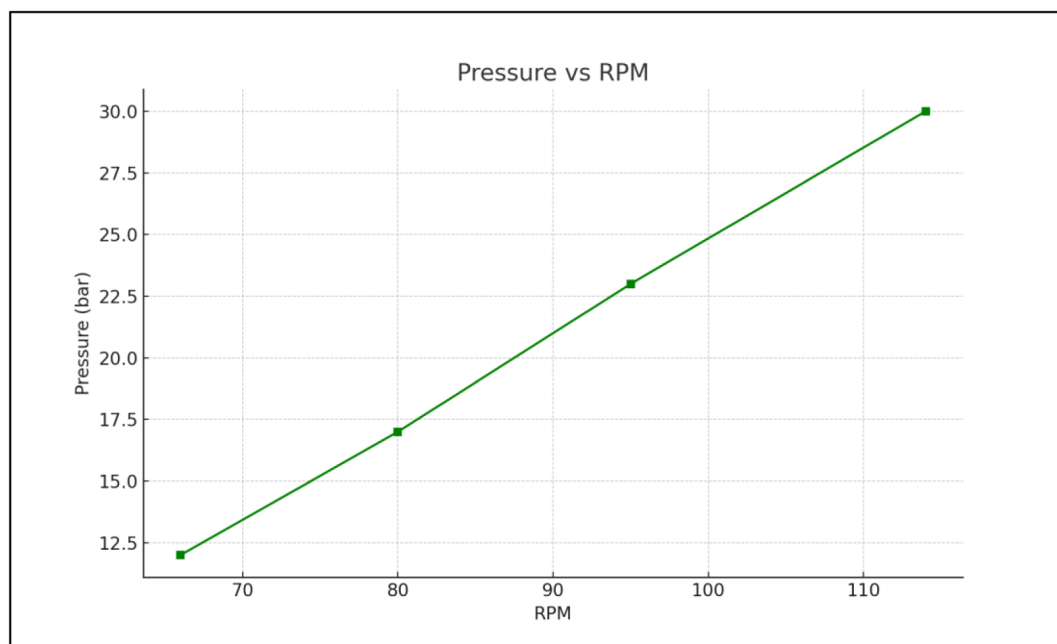


Figure 5—Pressure vs rotary speed relationship. Illustration of tool operating pressure as function of rotary speed under load.

Field Deployment: Vertical Section Deployment

Background

A vertical extended 12-in openhole interval in heterogeneous formations ranging from limestone to abrasive siltstone/sandstone zones presented challenges for 9 5/8" casing deployment, requiring an extended 3-4 days wiper trip prior to deploy the casing string. Swelling shales and instable sands required a robust casing delivery solution. Drilling with controlling instantaneous rate of penetration (ROP) is required due to the tendency of building inclination. Cavings are observed also while drilling requiring increasing in mud weight and forces pumping out/backreaming to get through.

Level 1 CwD technology is usually deployed to ensure covering the openhole with the casing. As an alternative, CRSPDPM tool was presented as an alternative to reduce the associated risks of rotating the long casing string. Previous attempts using level-1 CwD and rotating casing resulted in high surface torque reaching almost the make-up connection of the premium connection (26 kft·lbf). The centralization is affected by completely removing the centralizers in the openhole section to ease rotating the casing string.

Solution Implementation

The 9-5/8" CRSPDPM tool was installed at the bottom of the casing string, with safety sub mounted above. The openhole centralization was preserved as planned in the drilling program. Drilling mud properties are highlighted in [table 1](#). The maximum recommended solids content prior to running was specified to be maximized at 8% to not result in excessive wear and tear on shoe parts and lead to shorter operational life. However, the solids content of the actual mud was 15%. The tool was deployed despite this limitation. The system had 45 lb/bbls of medium and fine bridging materials to minimize the chance of lost circulation. The

recipe is shown below. The safety sub was not activated as the 15% solids content and bridging materials concentration did not affect the functionality of the tool.

- 15 ppb Soluflake medium.
- 15 ppb G seal fine.
- 15 ppb Fiber medium.

Table 1—Drilling mud properties in the vertical section deployment case.

Mud type	Polymer Mud
Mud weight, PCF	85
Filtrate API	2.9
Mud Cake	1
Nacl	167000 mg/L
% SOLIDS	15%
Fun. Visc. (s/qt):	66 CzJxF
VISC 600/300	88/62
PH	9.7 CzJxF

Surface test for reamer shoe was performed while assembling the BHA, below are the parameters. The reamer shoe was confirmed working normally as expected with stable pressure, witnessed by DSV and decision made to proceed with casing string RIH.

The tool successfully reamed through three major tight spots especially close to the bottom of the section. A flow rate of 100 gal/min was enough to help push the casing through the tight spots. The results are summarized in [table 3](#). To complement [table 3](#), [figure 6](#) illustrates reaming down through one of the tight holes. The CRSPDPM was pumped down with 100 gal/min flow rate through the tight spot. The hookload reduce from 437 klbs prior to washing down the casing string to about 370 klbs.

Table 2—Surface test differential pressure output in the vertical section deployment case.

Flow Rate, GPM	Pressure, PSI
50 gpm CzJxF	140 psi
75 gpm	160 psi CzJxF

Table 3—The reaming down parameters through the tight spots in the vertical section deployment case.

Flow Rate, GPM	Pressure, PSI	Down weight, KLb	Up weight, KLb
470	660 - 530	239	241
100	900	390	450
100	900	390	450

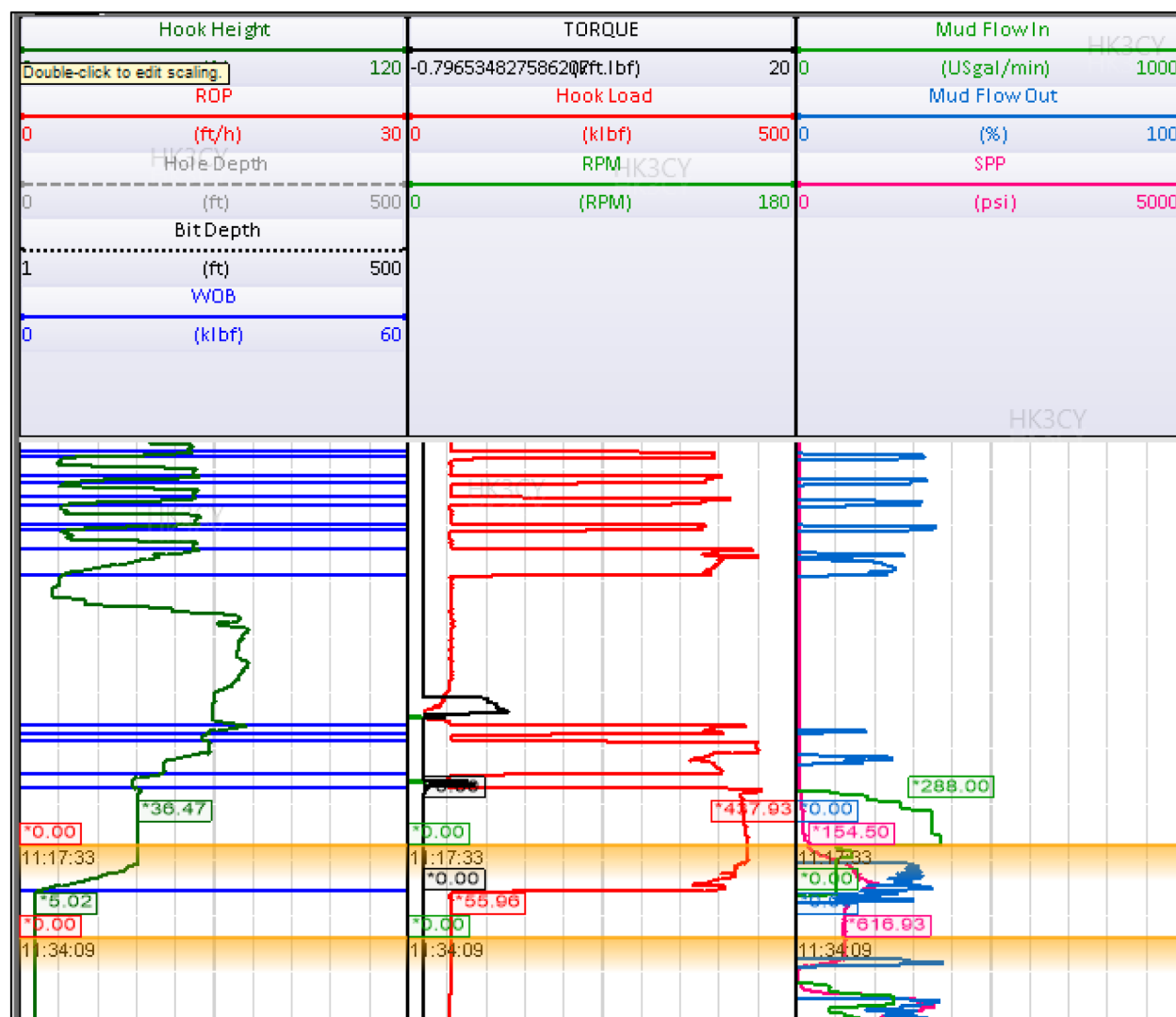


Figure 6—Real-time graph for washing down the 9-5/8-in CRSPD through a tight spot with 100 gal/min flow rate.

The casing string was cemented in place without issues. The overall pumping pressures while cementing followed the expected regime. Did not observe any spikes in pressures related to plugging of the CRMPD tool. The pumping pressures while cementing is summarized in table 4.

Table 4—The pumping pressures while cementing in the vertical section deployment case.

Flow Rate, GPM	Pressure, psi
256.2	550
256.2	700
210	400
489	320
489	800
489	1000
330	1800
146	1500

No joints damaged or rejected while RIH complete casing string. There was a problem with pipe cat and spider slips.

- Time spent to run the casing string to section TD was 25 hours. This is compared to about 3 days of wiper trip time.
- Time spent for circulation prior to cementing was 2 hours.
- Time spent for cementing the casing string was 4 hours.

Results

- Casing reached the section TD without casing rotation or the need to perform a lengthy wiper trip prior to deploying the casing string.
- The CRSPPDM tool was successfully ran and cemented without rotating the casing string. It is recommended to adopt regular practice of running the CRSPPDM as contingency tool in the wells to mitigate the risks of not delivering casing to the target depth, avoiding casing stuck incidents, and pulling the casing string out of the wellbore for dedicated reaming runs especially in cases where tight spots and unstable hole intervals anticipated.
- The CRSPPDM tool utilization can potentially result in saving the complete well or the drilled section. It is also confirmed to save the preparation wiper trip time prior to running the casing string, as well as reduced damage to casing connections while working on tight spots.
- The CRSPPDM tool utilization can be also considered for the task of saving additional casing string, when longer well interval can be completed with same casing size instead of drilling smaller size hole to accommodate smaller casing string.

Drill-out Lessons Learned

The drill-out process required two runs, which is an abnormal situation. The main reason for the need of a second run was the excessive WOB that was applied while drilling the CRSPPDM in combination with low surface RPM with PDC bit type. This bit types requires higher surface rotary speed for efficient drill-out. According to [figure 8](#), the applied WOB was increased to 47,000 lbs within the first 5-7 mins. The maximum recommended WOB to drill- out the CRSPPDM for 8- $\frac{3}{8}$ -in bit size is 22,000 lbs. Additionally, there were no bit pickups done while cleaning out to improve hole cleaning. As the WOB increased, the tangential component of the torque transmitted to the internal components of the equipment also increased, and at some point, in time, it exceeded the frictional force between the internal components and the housing of the shoe and cement around. This resulted in the failure of the component assembly, leading to the rotation of the internal parts. The bit started to rotate together with the internal sleeve, which prevented further progress in the drill out operations.

- Drill-out parameters applied in the first run:
 - WOB: 15,000 – 47,000 lbs
 - Torque: 3,000 – 9,000 ft-lbs
 - Surface rotary speed: 30 – 60 RPM.
 - Flow rate: 503 gal/min.
- Drill-out parameters applied in the second run:
 - WOB: 15,000 – 30,000 lbs
 - Torque: 3,000 – 6,000 ft-lbs

- Surface rotary speed: 15 – 70 RPM.
- Flow rate: 200-250 gal/min.

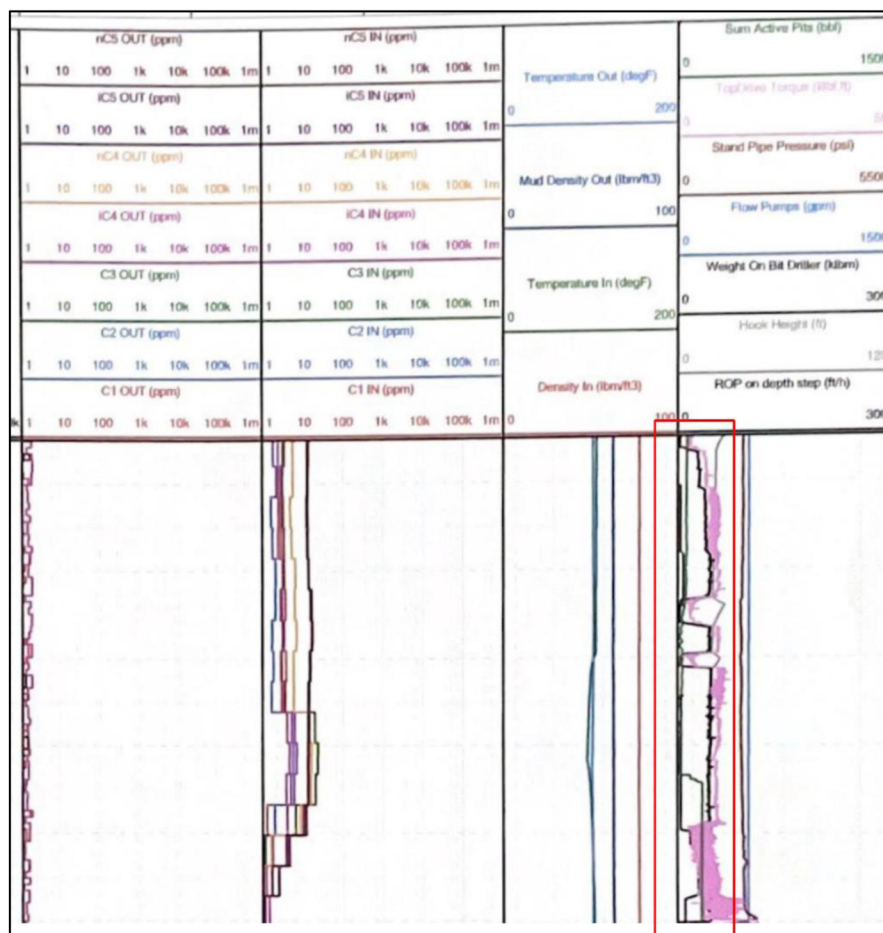


Figure 7—CRSPDPM drill-out real-time parameters. The highlighted WOB line in black shows that it was increased to an excessive value within the first few minutes of the drill-out process. Patience is required.

Field Deployment: Horizontal Section Deployment

Background

A horizontal 12-1/4-in openhole interval in a mature carbonate formation presented challenges for 9 5/8-in casing deployment. Swelling shale, ledges at 10,007 ft, and instability from depleted zones required a robust casing delivery solution. Previous attempts using float shoes and rotating casing resulted in high surface torque (28 kft-lbf) and connection galling.

Solution Implementation

The 9-5/8" CRSPDPM tool was installed at the bottom of the casing string, with safety sub mounted above. Drilling mud was programmed at 1.5 SG with 3% KCl and 3% LCM concentration. At TD, 2,900 LPM flow was established, generating tool downhole rotary speed of 110 RPM and torque output near 5,800 N·m.

The tool successfully reamed through three tight spots with less than 2 metric tons of hookload increase. Cementing was executed via standard shoe nozzles. Safety Sub was not activated.

Results

- Casing reached the section target depth (TD) without casing rotation or the need to perform a lengthy wiper trip prior to deploying the casing string.
- Drag reduced by 35% compared to offset wells as shown in [figure 8](#).
- Zero nonproductive time (NPT) from mechanical interference.
- Full cement returns achieved with no tool bypass activation.

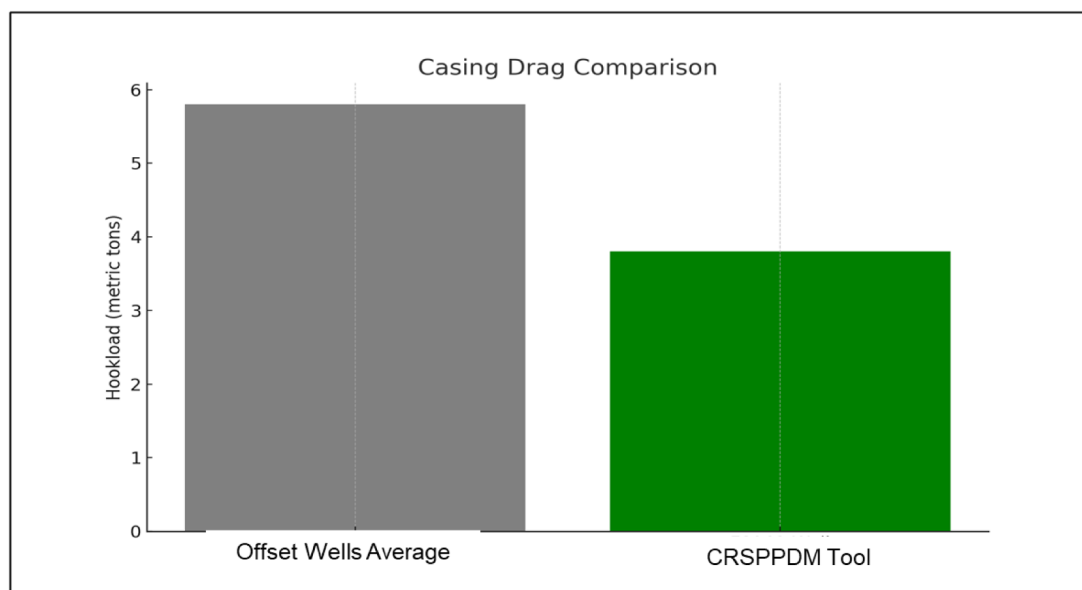


Figure 8—Casing drag comparison – comparison of hookload between traditional offset well and after CRSPDMD deployment. It shows ~35% reduction in casing drag.

Novelty and Differentiation

The CRMPPDM technology offers a novel and differentiated solution for casing deployment in extended open-hole sections. The technology's unique combination of a positive displacement motor and reaming cutters provides a reliable and efficient means of reaming through tight spots and collapsed walls, without the need for full-string rotation. This approach differs significantly from conventional casing deployment methods, which often rely on wiper trips, reaming-while-drilling (RWD), or rotating casing strings using top drives.

One of the key differentiators of the CRMPPDM technology is its ability to deliver localized high-torque, low-speed rotation at the casing shoe level. This allows for precise control over the reaming process, enabling the tool to effectively navigate through challenging wellbore conditions. In contrast, conventional casing deployment methods often rely on the complete rotation of the casing string, which can lead to increased torque and vibration, potentially causing damage to the casing string or wellbore.

The CRMPPDM technology also offers improved safety and reliability compared to conventional casing deployment methods. The tool's positive displacement motor provides a consistent and controlled source of power, reducing the risk of sudden torque spikes or vibrations that can cause damage to the casing string or wellbore. Additionally, the tool's reaming cutters are designed to be durable and long-lasting, minimizing the need for frequent replacement or maintenance.

In terms of operational efficiency, the CRMPPDM technology offers significant advantages over conventional casing deployment methods. The tool's ability to ream through tight spots and collapsed walls

without the need for full-string rotation reduces the time and effort required for casing deployment. This, in turn, can lead to significant cost savings and improved project economics.

Key Differentiators

- Localized high-torque, low-speed rotation: The CRMPPDM technology delivers precise control over the reaming process, enabling effective navigation through challenging wellbore conditions.
- Improved safety and reliability: The tool's positive displacement motor provides a consistent and controlled source of power, reducing the risk of sudden torque spikes or vibrations.
- Operational efficiency: The tool's ability to ream through tight spots and collapsed walls without full-string rotation reduces time and effort required for casing deployment.

Conclusion

The CRMPPDM technology is a game-changer, providing a reliable and efficient solution for casing deployment in extended open-hole sections. The technology has been tested in both laboratory and field settings, demonstrating its ability to maintain flow paths, deliver high torque outputs, and reduce casing drag and deployment time. The implementation of the CRMPPDM technology has improved operational efficiency, lowered casing failure risks, and ensured reliable bottomhole casing placement in extended lateral environments. The economic implications of the CRMPPDM technology are significant, as it reduces non-productive time, minimizes the risk of casing failure, and decreases the cost of casing deployment. The technology has been shown to be effective in a wide range of wellbore conditions, including high-pressure and high-temperature environments. The CRMPPDM technology has the potential to revolutionize the oil and gas industry by providing a reliable and efficient solution for casing deployment in challenging wellbores. As the oil and gas industry continues to evolve, the CRMPPDM technology is poised to play a critical role in the development of extended reach drilling and horizontal development wells. The technology has been designed to provide a reliable and efficient solution for casing deployment in extended open-hole sections, and its implementation has shown promising results in reducing the risks associated with conventional casing deployment methods. The CRMPPDM technology is a significant advancement in the field of casing deployment, and its potential to improve operational efficiency and reduce costs is substantial.

Abbreviations

- BHA : Bottom Hole Assembly
- CwD : Casing-while-Drilling
 - DV : Differential Valve
- GPM : Gallons per Minute
- KCl : Potassium Chloride
- LCM : Lost Circulation Material
- NPT : Non-Productive Time
 - PSI : Bounds per In^{ches Squared}
- ROP : Rate of Penetration
- RPM : Revolutions Per Minute
 - TD : Target Depth
- WOB : Weight on Bit